

CHAINS & DEBUGGING CHEATSHEET

AI Engineer Ready · Week 3 — Wiring Steps Together and Fixing Them When They Break

Golden Rule → output of step N becomes input of step N+1 → print between every step

1. What Is a "Chain"?

A chain is just: **the output of one step becomes the input of the next**. You don't need a special object to start — plain Python functions calling each other in sequence *are* a chain.

2. Pattern: Translate → Summarize

```
def call_llm(prompt_text):
    try:
        resp = chat.invoke(prompt_text)
        return getattr(resp, "content", str(resp))
    except Exception as e:
        return f"LLM error: {e}"

def translate(text, source_lang="English", target_lang="Spanish"):
    p = translate_template.format(text=text, source_lang=source_lang, target_lang=target_lang)
    return call_llm(p)

def summarize(text):
    return call_llm(summarize_template.format(text=text))

# chain them: output of translate() becomes input of summarize()
translated = translate("LangChain makes it easier to build reliable LLM apps.")
summary = summarize(translated)
```

Why wrap calls in `call_llm()`? One place for try/except, one place to swap models or add retries/logging, and every step function stays short.

3. Extending a Chain (e.g. Add a `tone` Parameter)

```
summarize_template = "Summarize in one paragraph, in a {tone} tone:\n\n{text}\n"

def summarize(text, tone="neutral"):
    return call_llm(summarize_template.format(text=text, tone=tone))

summarize(translated, tone="casual")
```

Pattern: add the new parameter to both the function signature and the template's placeholders, with a sensible default so existing calls don't break.

4. The 4-Step Debugging Process

STEP	ASK
1. Identify	Where does it break — API key? Model call? JSON parsing? Which cell?
2. Print	Intermediate values after every step, not just the final output

STEP	ASK
3. Check assumptions	API key loaded? Model name spelled right (gpt-4o-mini, not gpt4o)? Cells run in order?
4. Search	Copy the last line of the traceback, Google it

5. Common Errors

ERROR	LIKELY CAUSE	FIX
<code>NameError: 'chat' not defined</code>	Ran a cell before initializing chat	Run setup cell first, then re-run in order
<code>AuthenticationError</code>	Missing/invalid API key	Check <code>.env</code> , confirm <code>load_dotenv()</code> printed True
<code>KeyError</code> in <code>.format()</code>	Template placeholder not supplied	Pass every <code>{name}</code> used in the template
Chain output wrong, no error	Earlier step returned an unexpected value	Print the return value of every step
<code>RateLimitError</code>	Too many calls too fast / quota hit	Add delay between calls, check billing dashboard

Pre-Flight Checklist

- ☐ `.env` has `OPENAI_API_KEY=sk-...` (no quotes), in the right folder
- ☐ `load_dotenv()` ran and printed True
- ☐ Model name spelled correctly
- ☐ Raw response printed before any parsing
- ☐ Every API call wrapped in `try/except`
- ☐ Cells run top-to-bottom, in order